

HACKARENA 2.0 – HYDERABAD ZONALS | APPLIED AI TRACK

MediAgent

Agentic AI for Smarter Hospital
Operations & Electronic Health Record
Management

Eliminating Administrative Friction. Empowering Clinical Decisions.

Team: Jesse We Need To Cook

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AGENTIC AI PIPELINE



India's Healthcare Gaps

1:1456

Doctor-to-Patient

WHO recommends 1:1000

40%

Day Lost

Spent on documentation

6B

Visits/Year

Administrative overhead



Fragmented Records

Paper files or disconnected software—no unified history across departments.



Language Barriers

English-only systems exclude Telugu and Hindi-speaking patients.



Medication Error Risk

No pre-prescription conflict detection at the point of care.



No Intelligent Triage

Scheduling systems assign no priority based on clinical severity.

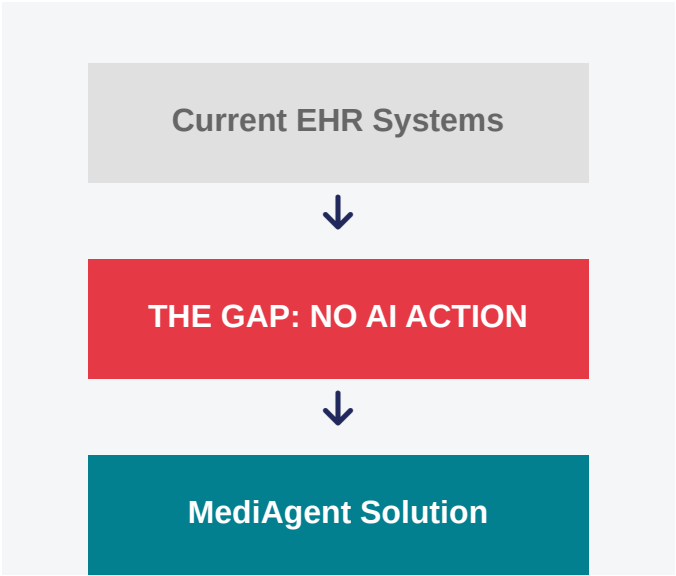
DROWNING IN PAPERWORK

THE REALITY OF CLINICAL ADMINISTRATION

Current EHR Systems Fall Short

Practo, HealthPlix, and ABDM cover 70%+ of digitised clinics in India but share fundamental limitations that MediAgent addresses.

Feature	Current Systems	MediAgent
AI Integration	No LLM or Agentic AI	Full Agentic Pipeline
Transcription	Basic digital storage	Real-time Voice AI
Language Support	English-only	Telugu & Hindi
Data Entry	Manual form-based	Auto-populated EHR
Safety Checks	Pharmacy-stage only	Pre-prescription Alert
Triage Scoring	None (Manual)	Intelligent Severity



Every step in existing systems requires manual input—no system acts autonomously. **MediAgent fills this gap entirely.**

Three AI Agents. One Seamless Workflow.

MediAgent introduces three specialised AI agents working together as one continuous automated pipeline.



1. Patient Agent

Multilingual conversational intake (English, Telugu, Hindi). Collects symptoms, history, and medications. Assigns severity scores (1-5).

PRE-CONSULTATION



2. Consultation Agent

Real-time voice transcription via OpenAI Whisper. Automated extraction of diagnosis, medications, and instructions using Claude API.

DURING CONSULTATION



3. EHR Agent

Automated drug-drug and drug-allergy conflict checks. Generates discharge summaries in regional languages and insurance documentation.

POST-CONSULTATION

"AI eliminates administrative friction without replacing clinical judgement. Every AI-generated output is a draft presented to the doctor for review."



How Each Agent Works — Step by Step

Agent	Trigger	Step-by-Step Actions	Final Output
 Patient Agent	Booking initiated on portal	Multilingual intake (Eng/Tel/Hin) → Symptom & history collection → Emergency keyword detection → Severity scoring (1-5)	Structured PDF Pre-Consultation Report
 Consultation Agent	Doctor opens patient case	Audio consent → Real-time streaming → Whisper transcription → Claude extraction (Diagnosis, Meds, ICD-10)	Auto-populated Editable EHR Draft
 EHR Agent	Doctor clicks "Approve"	DrugBank conflict checks → Record finalization → Multilingual discharge docs → Insurance & Reminders	Immutable EHR + Patient Docs



Key Rule: No AI output becomes a final medical record without explicit doctor review and approval. Human oversight is built into every stage.

Built for Real Clinical Environments

PATIENT PORTAL



Multilingual Conversational Intake

Natural chat in English, Telugu, and Hindi. No forms to fill.



Emergency Auto-Flagging

Instant escalation to Severity 5 for critical symptoms like chest pain.



Treatment Timeline View

Complete medical history and lab results in a single accessible screen.



Document Download

Prescriptions and discharge summaries available as PDF anytime.



MOBILE INTERFACE

DOCTOR DASHBOARD



AI Priority Queue

Patients sorted by AI-assigned severity score (1–5) for urgent care.



Full EHR in One Screen

Unified history without switching between modules or departments.



Drug Conflict Alerts

Automated checks against allergy list and current medications.



One-Click Insurance Generator

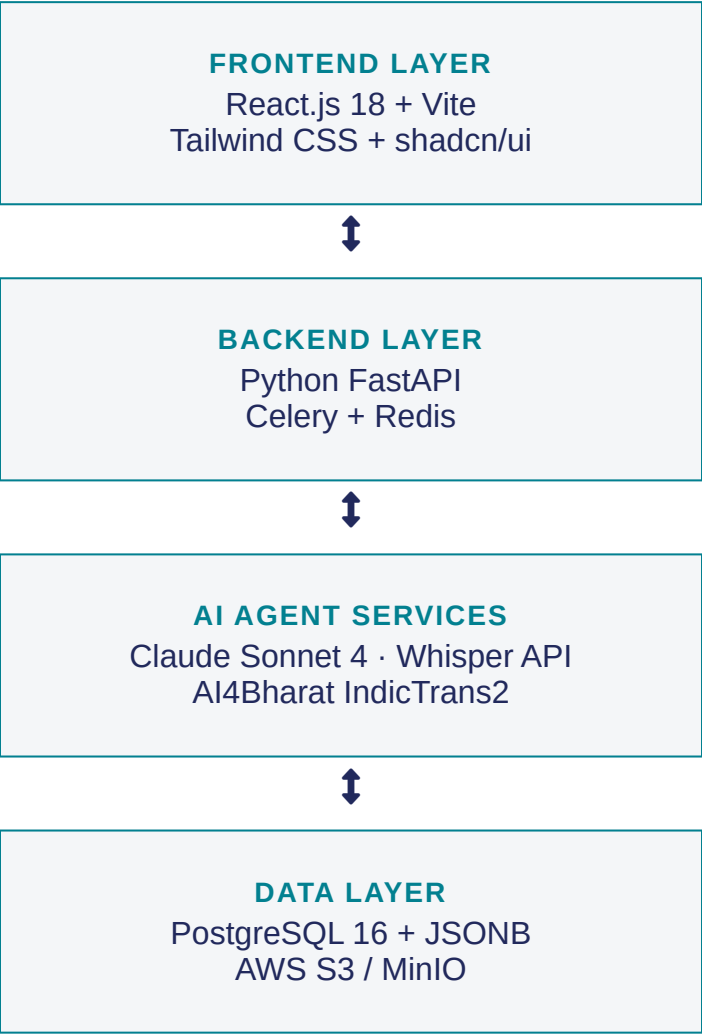
Auto-generated standard documentation from approved records.



CLINICAL WORKSPACE

Pragmatic Stack, Production-Ready Design

Selected for rapid prototyping while remaining fully production-scalable.



Technology	Why This Choice?
Anthropic Claude 4	Best-in-class for multilingual reasoning and structured JSON output.
OpenAI Whisper	High accuracy speech-to-text with robust code-switching support.
Python FastAPI	High-performance async support for concurrent AI agent calls.
PostgreSQL + JSONB	ACID compliance with flexible storage for complex EHR data.
IndicTrans2	State-of-the-art translation for English ↔ Telugu/Hindi.
Docker	Ensures consistent development and production environments.

Safe, Compliant, and Built to Scale

1 BILLION +

Hours of clinical productivity restored annually across India

PROTECTED BY DESIGN



Encryption at Rest & Transit

AES-256 for PII fields and TLS 1.3 for all API/WebSocket communications.



DPDP Act 2023 Compliant

Explicit user consent, role-based access control, and data deletion rights.



Immutable Audit Logging

Append-only logs for every AI action, doctor edit, and record approval.

IMPACT AT SCALE



~2 Hours Saved Per Doctor

Massive increase in clinical capacity without hiring new medical staff.



Healthcare in Your Language

Closing accessibility gaps for Telugu and Hindi-speaking populations.



Faster Reimbursements

Auto-generated insurance docs reduce claim rejections and improve cash flow.

Our Plan to Build & Demonstrate

PHASE 1: 8-HOUR HACKATHON BUILD

- ✓ Multilingual Patient Agent Intake
- ✓ Pre-Consultation PDF Generation
- ✓ EHR Draft Review Workflow
- ✓ AI Severity Scoring (1-5)
- ✓ Doctor Dashboard Priority Queue

PHASE 2: POST-HACKATHON EXTENSIONS

- 🕒 Live Voice Transcription (Whisper)
- 🕒 Regional Discharge Summaries
- 🕒 SMS/Email Follow-up System
- 🕒 DrugBank Conflict Detection
- 🕒 Auto Insurance Documentation

MediAgent

"Augmenting Doctors.
Not Replacing Them."

MediAgent handles the paperwork so doctors can focus on patients.

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